

# Roshan Hegde

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## EDUCATION

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### University of Illinois at Urbana-Champaign

May 2027

*B.S. in Computer Science, Grainger College of Engineering*

*GPA: 3.91/4.0*

*B.S. in Mathematics, College of Liberal Arts & Sciences*

### Highland Park High School

May 2023

*GPA: 4.69/4.0*

## RELEVANT COURSEWORK

**Courses:** Algorithms, Operating Systems, Computer Architecture, Programming Languages and Compilers, Cryptography, Data Structures, Probability and Statistics for Computer Science, Applied Linear Algebra

## SKILLS

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**Languages:** C, C++, Rust, Go, Java, Kotlin, Python, Typescript

**Tools:** Git, Docker, Linux, Nix, Vim

## WORK EXPERIENCE

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### Software Engineering Intern, Chicago IL | *Olam Food and Ingredients* | *Azure, python*

May – August 2025

- Analyzed current utilization of cloud assets and identified opportunities to improve cost efficiency by 12%
- Developed a data analysis service to automate VM performance reporting to identify over-allocated resources
- Proposed a plan to improve resource cost-effectiveness to the infrastructure team

### University Of Illinois Urbana-Champaign | *Operating Systems Course Assistant* | *C, GDB*

August 2025 - Present

- Developed and reviewed programming assignments for students taking the course
- Answered both conceptual and implementation questions while hosting office hours

### MechMania, Champaign IL | *Lead Infrastructure Developer* | *Go, Rust, Python, Nix*

August 2024 – Present

- Built a multithreaded web-socket server in Go interacting with AWS services including databases and cloud storage
- Implemented a custom shared memory IPC protocol for multi-language support
- Leveraged Nix and bubble-wrap to run arbitrary code without vulnerabilities

## PROJECTS

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### LumonOS: Operating System | *C, Unix, RISC-V, QEMU, GDB*

January – August 2025

- Created fully functional, RISC-V, Unix-based pre-emptive operating system with round robin scheduling policy.
- Built a virtio block driver, cryptography driver, and entropy driver
- Implemented support for pipes between processes and argument passing according to POSIX standard
- Placed 3rd in the Operating System design competition hosted by the ECE department

### nix-tinker | *Rust, Cargo, Nix*

May 2024 – Present

- Developed an open-source tool for rapid prototyping of Nix configuration files, streamlining development workflow
- Crafted an intuitive CLI for managing file links with the Nix store, significantly enhancing user efficiency
- Leveraged Nix Flakes to guarantee reproducible and consistent build environments across diverse systems

## AWARDS

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### 1st place, Hackillinois '24, '25 | *Rust, Cargo*

February 2024 and 2025

- Competed with over 1,100 other students at Illinois' largest Hackathon
- Developed a full stack developer tool intended for rapid prototyping of data pipelines
- Defined a strongly typed JSON based protocol for ease of extensibility and versatility
- Built a developer tool used to universally initialize a basic development environment for over 30 languages

### 1st place, MechMania '24, '25 | *Java, Python*

September 2023 and 2024

- Implemented A\* pathfinding with a modified Manhattan distance heuristic
- Fine-tuned heuristics to pilot dogfighting planes, leveraging predictive modeling to gain a competitive edge